

Article

Performance of a Novel Computational Hyperemic Resistance Index Derived from Cardiac CT in Coronary Chronic Syndromes

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Abstract: Background/Objectives: Coronary artery disease (CAD) remains the leading global cause of mortality, underscoring the need for functional assessments that extend beyond anatomical evaluation. The Hyperemic Stenosis

Resistance (HSR) index combines invasive pressure and flow parameters to assess stenosis severity but faces limitations due to methodological and standardization challenges. This study aimed to introduce and validate a novel non-invasive computational equivalent of HSR (cHSR), derived from coronary computed tomography angiography (CCTA), and to compare its diagnostic performance with fractional flow reserve derived from computational fluid dynamics (FFR_{CFD}) and quantitative flow ratio (QFR). **Methods:** A retrospective analysis was conducted on 64 patients (106 coronary lesions) with suspected chronic coronary syndrome (CCS) who underwent both CCTA and invasive coronary angiography (ICA). Computational simulations incorporated patient-specific boundary conditions based on CCTA-derived left ventricular and aortic flow data. Diagnostic accuracy for predicting revascularization was compared among cHSR, FFR_{CFD}, and QFR. **Results:** FFR_{CFD} showed a strong correlation with invasive FFR ($r = 0.87$, $p < 0.0001$). The cHSR index achieved the highest diagnostic accuracy (96.2%) at an optimal cut-off of 0.75 mmHg/cm·s⁻¹, outperforming both FFR_{CFD} and QFR. No significant correlation was found between cHSR and shear stress parameters, including the Oscillatory Shear Index (OSI) and Time-Averaged Wall Shear Stress (TAWSS), indicating complex hemodynamic interactions beyond simple flow–pressure relationships. **Conclusions:** The computational

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hyperemic stenosis resistance (cHSR) index represents a promising non-invasive tool for the functional assessment of CAD, demonstrating superior diagnostic performance compared with existing imaging-based indices. Prospective multicenter studies with larger populations are warranted to confirm its clinical applicability and prognostic value in chronic coronary syndrome management.

Keywords: computational fluid dynamics (CFD); coronary computed tomography angiography (CCTA); hyperemic stenosis resistance (HSR); fractional flow reserve (FFR); chronic coronary syndrome (CCS); wall shear stress (WSS)

1. Introduction

Coronary artery disease (CAD) is the primary global cause of mortality, accounting for 9 million deaths annually, or roughly one in seven deaths worldwide, according to the World Health Organization (WHO). While invasive coronary angiography (ICA) and quantitative coronary angiography (QCA) traditionally assess the anatomical severity of stenosis, relying solely on these measures often proves insufficient. In different series, 20–30% of lesions deemed severe by QCA do not induce ischemia during functional testing [1]. This discrepancy highlights the crucial need for integrating functional evaluations into the revascularization decision-making process to improve patient outcomes [2].

Fractional Flow Reserve (FFR) has emerged as the gold standard for functional lesion assessment and is strongly predictive of adverse cardiovascular events [3–5]. Despite its robustness, FFR does not provide information about the microvascular function, with a known discrepancy between FFR and Coronary Flow Reserve (CFR) in up to 20% cases in both normal and altered CFR populations [6]. While FFR assesses pressure gradients across stenosis under hyperemia, microvascular resistance is neutralized by hyperemic agents (adenosine mainly). However, cases of microvascular dysfunction can cause ischemia independently of epicardial stenosis [7,8]. To address these gaps, the Hyperemic Stenosis Resistance (HSR) index was introduced, combining pressure and flow data to offer a more refined physiological assessment [9]. However, the lack of standardized methods for flow measurement has limited its routine clinical use [10]. In a recent study conducted by Boerhout and al., using data from 853 patients and 1107 vessels from the multicentric ILIAS Registry, they aimed to assess HSR's diagnostic and prognostic value in chronic coronary syndrome (CCS) patients. HSR more accurately identified inducible ischemia compared to FFR and CFR (AUC 0.71 vs. 0.66 and 0.62, respectively; $p < 0.005$). Additionally, abnormal HSR was a strong, independent predictor of target vessel failure

at 5-year follow-up (HR 3.80, 95% CI: 2.12–6.73; $p < 0.005$). In cases where revascularization was deferred, HSR showed superior ability in pinpointing vessels that might still benefit from intervention [11].

In the quest for non-invasive diagnostic alternatives, functional imaging techniques like stress SPECT, echocardiography, and cardiac magnetic resonance (CMR) are commonly used to evaluate ischemia [2]. However, their diagnostic accuracy can be limited in high-risk populations, such as patients with left main or three-vessel disease, where anatomical assessment is crucial for risk stratification [12]. In such cases, combining functional and anatomical data is necessary and may provide a more comprehensive evaluation.

Recent advancements in computational modeling, based on Computational Fluid Dynamics (CFD), have enhanced the diagnostic utility of anatomical imaging by simulating coronary flow and pressure [13]. Quantitative Flow Ratio (QFR), based on ICA, and Computed Tomography Fractional Flow Reserve (FFR_{CT}), based on coronary computed tomography angiography (CCTA), both use CFD models to estimate the functional significance of coronary stenoses [14,15]. These techniques have demonstrated improved diagnostic accuracy by offering both anatomical and functional insights with an acceptable cost-effectiveness ratio [16]. However, the performance of a model relies heavily on the defined boundary conditions (BCs). Most available commercial software uses theoretical assumptions and idealized parameters, which can miss the patient-specific setting. When clinical data are accessible, using patient-specific BCs is preferable over generalized or estimated parameters, which produce more realistic flow/pressure curves, particularly near side branches and curvatures where atherosclerotic plaques commonly form [17].

This study proposes a new non-invasive computational equivalent of Hyperemic Stenosis Resistance (cHSR) index derived from CCTA to provide a combined flow/pressure-based assessment of coronary lesions. We aim to evaluate its diagnostic performance in predicting revascularization, with FFR_{CFD} and QFR serving as comparators. Furthermore, we explored the relationship between cHSR and shear stress parameters, such as Oscillatory Shear Index (OSI) and Time-Averaged Wall Shear Stress (TAWSS).

2. Methods

2.1. Population of Study and Inclusion/Exclusion Criteria

This retrospective observational outcome-blinded study was conducted at Brest University Hospital between January 2021 and September 2024. Patients referred for

CCTA due to suspected chronic coronary syndrome (CCS), who subsequently underwent ICA, were included. The local Institutional Ethics Committee approved the study protocol, which was based on Reference Methodology (MR004), as defined by the French Data Protection Authority (CNIL), and patient data were anonymized in compliance with institutional regulations (Figure 1).

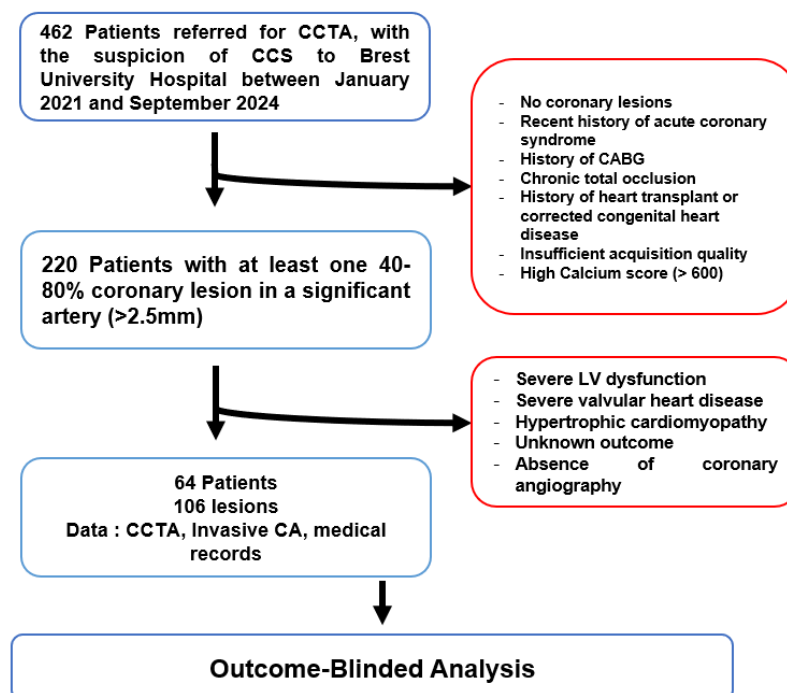


Figure 1. Study flowchart CCTA: Coronary Computed Tomography Angiography; CCS: ChronicCoronary Syndrome; CABG: Coronary Artery Bypass Grafting; LV: Left Ventricular; CA: Coronary Angiography.

Inclusion criteria were patients aged between 18 and 85 years who underwent both CCTA and ICA, with at least one significant lesion (40–80%) in a coronary artery larger than 2.5 mm in diameter. Patients were excluded if they had a recent acute coronary syndrome (<40 days), a left ventricular ejection fraction (LVEF) <40%, significant mitral or aortic valve disease, prior coronary artery bypass graft surgery (CABG), heart transplant, or the presence of chronic total occlusion (CTO). Sixty-four patients with 106 coronary lesions were included in the analysis; the final revascularization decision was omitted from the analysts throughout the study protocol.

In our cohort, the decision whether to proceed with revascularization or not was informed by various methods. Invasive fractional flow reserve (FFR) guided 60% ($n = 63$ lesions) of the revascularization decisions. An additional 30% ($n = 32$ lesions) of the decisions were based on stress-induced ischemia detected by stress single-photon emission computed tomography (SPECT), while in the remaining 10% ($n = 11$ lesions) of patients, revascularization was indicated by electrocardiographic changes and the presence of wall motion abnormalities on rest imaging. Clinical coronary disease pretest probability and coronary calcium scores were calculated in accordance with the most recent guidelines [2].

2.2. Data Acquisition

CCTA was performed using a 320-slice volumetric CT scanner (Aquilion ONE, Toshiba Medical Systems) with retrospective ECG-gating and tube modulation. Image acquisition parameters included a slice thickness of 0.6 mm, with a non-ionic contrast agent injected at 5 mL/s followed by a saline flush. The images were reconstructed using Slicer 3D software; segmentation and meshing of coronary artery geometries were conducted with SimVascular, an open-source cardiovascular modeling platform [18]. For each patient, we calculated total left ventricular myocardial mass and allocated percentages for each artery through visual analysis. Invasive blood pressure waveforms (which can be replaced by non-invasive blood pressure), stroke volume, and systolic ejection time were important input data for the sake of individualized flow and pressure targets in the parameters tuning process (Figure 2).

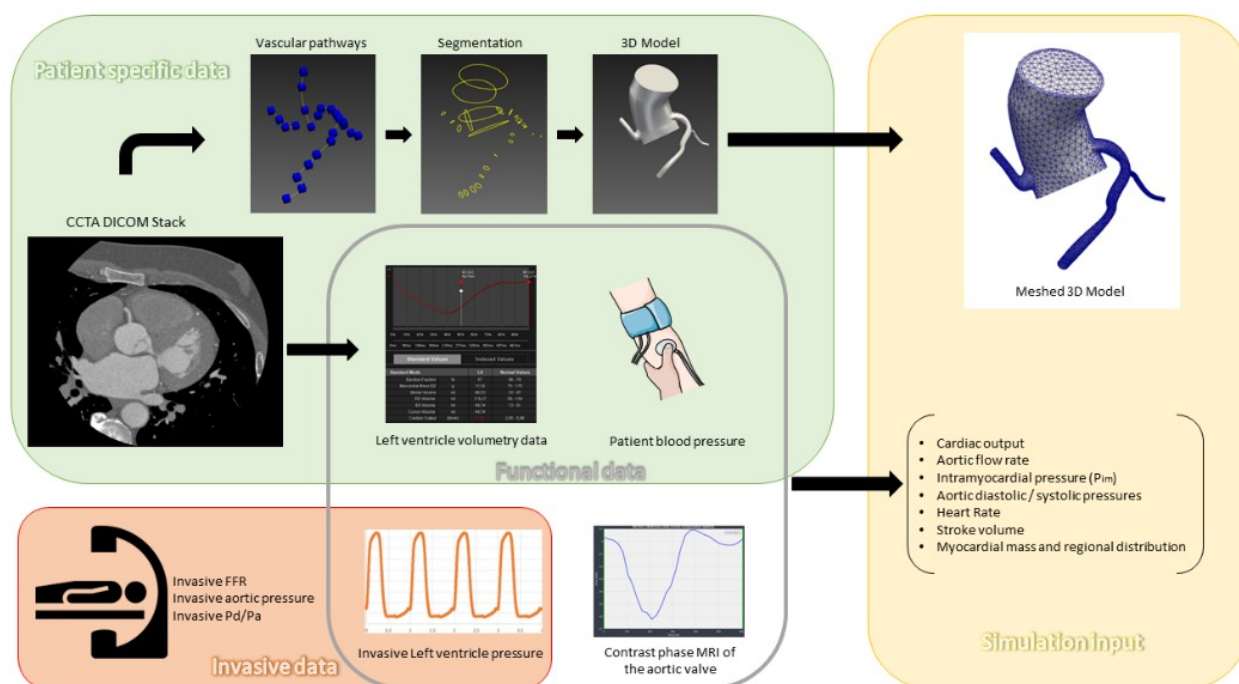


Figure 2. Patient data preprocessing: CCTA: Coronary Computed Tomography Angiography; DICOM: Digital; Imaging and Communications in Medicine; FFR: Fractional Flow Reserve; Pa: Aortic Pressure; Pd: Distal Coronary Pressure; MRI: Magnetic Resonance Imaging.

In addition, we modeled aortic flow rates using phase contrast cardiac MRI from 15 healthy individuals, adjusted according to each patient's body surface area (BSA). For intramyocardial pressure modeling, we acquired left ventricular pressure measurements from 22 healthy individuals, also scaled by BSA, to simulate patient-specific myocardial loading conditions more accurately.

ICA was performed following standard clinical protocols, and invasive FFR measurements were obtained using the PressureWire X (Abbott Vascular, Santa Clara, CA, USA). Hyperemia was induced using intravenous adenosine at 140 $\mu\text{g}/\text{kg}/\text{min}$ besides intracoronary nitroglycerin to avoid spastic coronary lesions. For the purpose of internal model validation, we collected pressure waveforms of the 60% of lesions that had invasive FFR measurements. Quantitative Flow Ratio (QFR) was derived from ICA images through QAngio XA 3D software (research edition, version 2.1, Medis, Leiden, The Netherlands).

2.3. Simulations Pipeline and Measurements

Our simulation pipeline used a patient-specific model (which consisted of the initial aorta, and the three coronary arteries) obtained from CCTA, with segmentation and meshing performed in SimVascular platform (as detailed in Supplementary Materials) [18]. Aortic inflow boundary conditions were calculated based on CCTA-derived left ventricular volumetric data to produce a patient-specific aortic flow rate curve (which guided scaling of curves obtained in MRI to adapt to each patient's hemodynamics and body surface area). Invasive aortic and intraventricular pressure measurements taken during the ICA served both to calculate initial resistances and as targets for lumped parameters tuning.

Outlet boundary conditions of the peripheral arterial system were based on a simple three-element Windkessel model containing resistances (R_p for proximal arteries and R_d for distal and microvascular systems) and a capacitor (aortic compliance) [19]. Coronary outlet boundary conditions were adapted from Mantero et al. [20] and modified to exclude coronary venous microcirculation compliance for computational simplicity while preserving realistic coronary flow and pressure dynamics. This coronary vascular model incorporated arterial resistance (R_a) and compliance (C_a), arterial microcirculation resistance ($R_{a-micro}$), myocardial compliance (C_{im}), venous microcirculation resistance ($R_{v-micro}$), and intramyocardial pressure (P_{im}) to achieve patient-specific hemodynamic responses.

To optimize computational accuracy, both models' lumped parameters were refined using reduced-order simulations, and different conditions (Rest and exercise) were tested according to the methods proposed by Kim and al. [21]. Initial values and tuning process are detailed in Supplementary Materials. Tridimensional CFD simulations were executed after that, using the SimVascular gateway [22], solving Navier-Stokes equations to compute hemodynamic parameters, including pressure, velocity, wall shear stress (WSS), and flow at each point of the coronary arteries tree. These simulations were performed under both basal and hyperemic conditions and produced output, allowing for assessment of stenosis severity in relation to functional coronary flow (Figures 3 and 4). FFR_{CFD} was calculated as the ratio of distal coronary pressure P_{dist} to aortic pressure P_{aortic} during hyperemia:

$$FFR_{CFD} = \frac{P_{dist}}{P_{aortic}}$$

Similarly, cHSR was calculated by dividing the pressure gradient across the lesion (ΔP) by the hyperemic average peak velocity (hAPV) as measured at stenosis level through 3D CFD simulation results:

$$cHSR = \frac{\Delta P}{hAPV}$$

Wall shear stress (WSS) quantifies the tangential force exerted by blood flow on the vessel wall and plays a critical role in endothelial function and atherosclerosis progression. Time-Averaged Wall Shear Stress (TAWSS) was computed using the equation:

$$TAWSS = \frac{1}{T} \int_0^T |\tau(t)| dt$$

where $\tau(t)$ represents the instantaneous WSS at a given point on the arterial wall, and T is the cardiac cycle duration. This provides an overall measure of shear stress magnitude experienced by the vessel wall over the cardiac cycle.

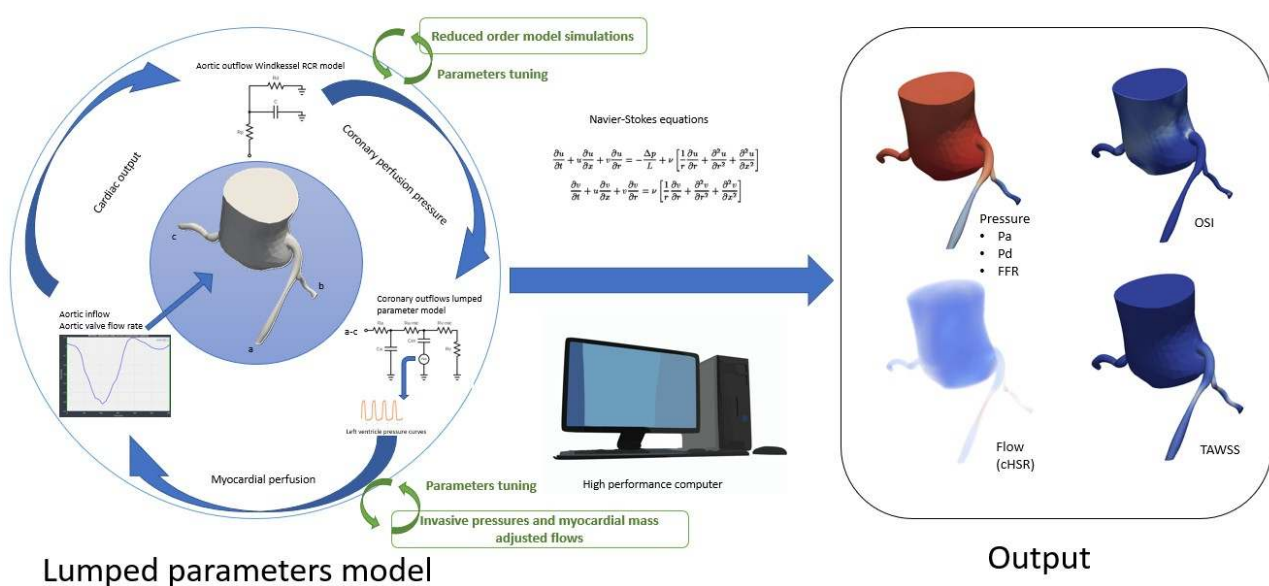


Figure 3. Simulation process and output: Pa: Aortic Pressure; Pd: Distal Coronary Pressure; FFR: Fractional Flow Reserve; cHSR: Computational Hyperemic Stenosis Resistance; OSI: Oscillatory Shear Index; TAWSS: Time-Averaged Wall Shear Stress; RCR Model: Resistance, Capacitance, and

Resistance model (used in Windkessel outflow modeling); Navier-Stokes Equations: Fundamental equations for fluid dynamics, used to model blood flow.

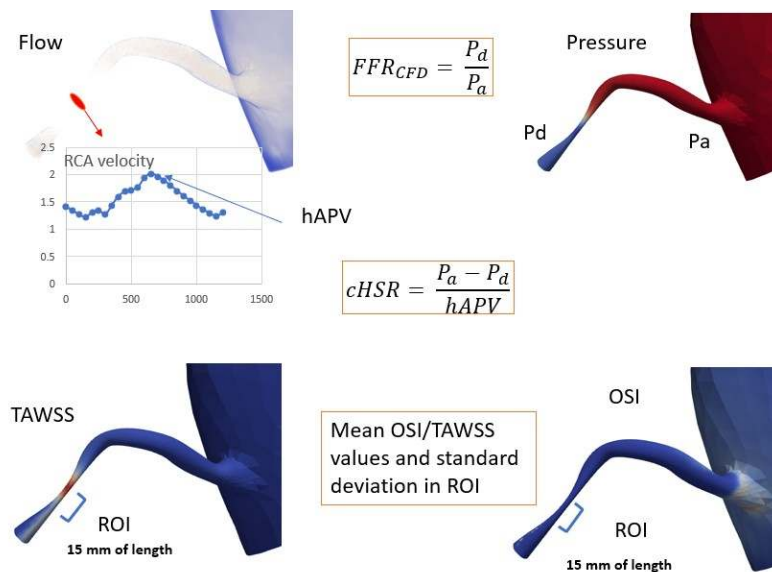


Figure 4. Computational analysis illustrating key parameters: FFR_{CFD} : Fractional Flow Reserve via Computational Fluid Dynamics; hAPV: Hyperemic Average Peak Velocity; cHSR: Computational Hemodynamic Stenosis Resistance; Pd: Distal Pressure, Pa: Proximal Pressure; TAWSS: Time-Averaged Wall Shear Stress; OSI: Oscillatory Shear Index; ROI: Region of Interest, 15 mm of vessel length beyond peak stenosis.

2.4. The Oscillatory Shear Index (OSI) Quantifies the Directional Variability of WSS and Is Calculated as

$$OSI = \frac{1}{2} \left(1 - \frac{\left| \int_0^T \tau(t) dt \right|}{\int_0^T |\tau(t)| dt} \right)$$

where the numerator accounts for the net directional shear stress over a full cycle, while the denominator represents the total shear magnitude. OSI values close to 0 indicate unidirectional flow, whereas values approaching 0.5 suggest significant oscillations in shear direction. Both TAWSS and OSI were computed from CFD simulations using

Navier-Stokes equations under pulsatile flow conditions, with coronary flow profiles extracted from patient-specific CCTA-derived aortic inflow and coronary geometry. These parameters were analyzed at regions of interest (15 mm beyond the peak stenosis) to evaluate their relationship with cHSR and lesion severity.

2.5. Statistical Analysis

For statistical analysis in our study, we used R Statistical Software (v4.4.1; R Core Team 2021). Descriptive statistics were used to summarize the baseline characteristics of the study population and coronary lesions. Normally distributed continuous variables were presented as mean and standard deviation (SD), while non-normally distributed variables were expressed as median and interquartile range (IQR). The Shapiro-Wilk test was used to assess normality, and categorical variables were described as counts and percentages.

The agreement between FFR_{CFD} and invasive FFR was assessed using a Bland-Altman plot, and their correlation was evaluated using Pearson's correlation coefficient. The optimal cut-off for cHSR was determined using Youden's index. Diagnostic performance of cHSR for predicting revascularization was assessed using ROC curve analysis, with area under the curve (AUC), sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) calculated for each method. All these metrics were compared to those of FFR_{CFD} and QFR using the DeLong test.

Correlation between cHSR and wall shear stress parameters (TAWSS, OSI) was evaluated using Pearson's correlation coefficient. Statistical significance was set at $p < 0.05$.

3. Results

3.1. Patient Population and Lesion Characteristics

A total of 64 patients (mean age 67.8 ± 9.2 years) were included in the study, with a pre-test probability for coronary artery disease (CAD) averaging $24.5 \pm 14.8\%$. The cohort had a sex ratio of 3:1. Among the patients, 62% had hypertension (HTN), 39% had diabetes mellitus (DM), 41% had a history of smoking, and 40% had dyslipidemia. Obesity was noted in 34% of the cohort, and 20% had chronic kidney disease (CKD). The mean coronary artery calcium (CAC) score was 113.1 ± 102.4 Agatston units (Table 1).

Table 1. Baseline Characteristics of the Study Population.

Characteristic	Mean $\pm$ (SD) or Percentage (%)
Total Patients	$n = 64$
Age (years)	67.8 ± 9.2
Sex Ratio	3 Male:1 Female

Hypertension (HTN)	62% (<i>n</i> = 40)
Diabetes Mellitus (DM)	39% (<i>n</i> = 25)
Smoking History	30% (<i>n</i> = 19)
Dyslipidemia	40% (<i>n</i> = 26)
Obesity	34% (<i>n</i> = 22)
Chronic Kidney Disease (CKD)	20% (<i>n</i> = 13)
Pre-test Probability	24.5 ± 14.8
Calcium Score (CAC) (Agatston)	113.1 ± 102.4

CAD: Coronary Artery Disease; HTN: Hypertension; DM: Diabetes Mellitus; CKD: Chronic Kidney Disease; CAC: Coronary Artery Calcium score.

In terms of coronary lesions, a total of 106 lesions were analyzed, with a mean stenosis length of 18.0 ± 7.4 mm. The distribution of lesions across different coronary arteries was as follows: 49% in the left anterior descending artery (LAD), 26% in the left circumflex artery (LCx), 22% in the right coronary artery (RCA), and 3% in the left main artery (LM). The mean stenosis percentage, as determined by quantitative coronary angiography (QCA), was $55.3 \pm 11.2\%$ (Table 2).

Table 2. Lesion Characteristics.

Lesion Location	Mean +/- (SD) or Percentage (%)
Total lesions	<i>n</i> = 106
LAD	51 (49%)
LCx	26 (26%)
RCA	23 (22%)
Left Main (LM)	3 (3%)
QCA (% Stenosis)	55.3 ± 11.2
Stenosis Length (mm)	18.0 ± 7.4

LAD: Left Anterior Descending artery; LCx: Left Circumflex artery; RCA: Right Coronary Artery; LM: Left Main artery; QCA: Quantitative Coronary Angiography.

3.2. Correlation Between FFR_{CFD} and Invasive FFR

A Bland-Altman analysis was used to assess the agreement between FFR_{CFD} and invasive FFR, revealing good agreement with a Pearson correlation coefficient of 0.87 ($p < 0.0001$), indicating strong linear correlation. The plot confirmed minimal bias (mean difference + 0.01), supporting the validity of the computational FFR_{CFD} measurements.

3.3. cHSR Cut-Off Determination

The optimal cut-off value for cHSR to predict the need for revascularization was determined through receiver operating characteristic (ROC) curve analysis and the Youden Index. The ideal cHSR cut-off was found to be 0.75 mmHg/cm·s⁻¹, providing the highest overall diagnostic accuracy. At this cut-off, the sensitivity and specificity for predicting revascularization were 96.2% each.

3.4. Diagnostic Performance of cHSR, FFR_{CFD}, and QFR

The diagnostic performance of cHSR, FFR_{CFD}, and QFR for predicting the necessity of revascularization was compared (Figure 5). cHSR demonstrated the best diagnostic accuracy (96.2%) compared to FFR_{CFD} (83.0%) and QFR (81.1%). cHSR also showed superior sensitivity and specificity (96.2%) when compared to FFR_{CFD} (sensitivity 71.7%, specificity 94.3%) and QFR (sensitivity 75.5%, specificity 86.8%). The positive predictive value (PPV) and negative predictive value (NPV) for cHSR were also both 96.6%, significantly higher than those of FFR_{CFD} and QFR (Table 3).

Table 3. Diagnostic Performance Metrics of cHSR, FFR_{CFD}, and QFR.

Metric	Sensitivity	Specificity	Accuracy	PPV	NPV	AUC
cHSR	96.20%	96.20%	96.20%	96.20%	96.20%	0.96
FFR _{CFD}	71.70%	94.30%	83.00%	92.70%	76.90%	0.93
QFR	75.50%	86.80%	81.10%	85.10%	77.90%	0.81

cHSR: Computational Hyperemic Stenosis Resistance; FFR_{CFD}: Fractional Flow Reserve computed from Computational Fluid Dynamics; QFR: Quantitative Flow Ratio; PPV: Positive Predictive Value; NPV: Negative Predictive Value; AUC: Area Under the Curve.

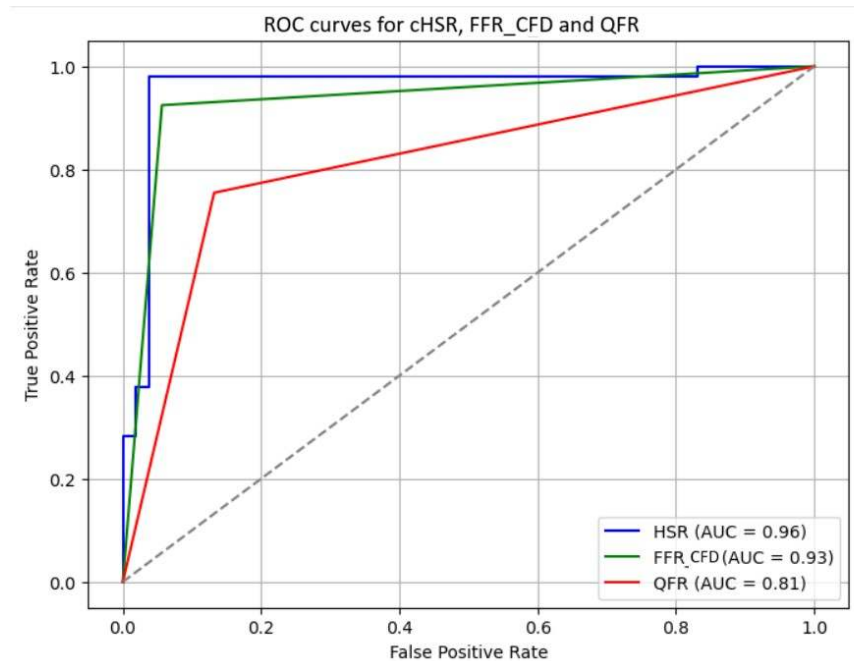


Figure 5. Receiver operator curves for cHSR, FFR_{CFD}, and QFR: cHSR: Computational Hyperemic Stenosis Resistance; FFR_{CFD}; Computed Fractional Flow Reserve; QFR: Quantitative Flow Ratio; AUC: Area Under the Curve; ROC: Receiver Operating Characteristic.

3.5. DeLong Test Results

The DeLong test was used to statistically compare the areas under the ROC curves (AUCs) of cHSR, FFR_{CFD}, and QFR. The results showed that cHSR had a significantly higher AUC compared to both FFR_{CFD} ($z = 19.27, p < 0.001$) and QFR ($z = 16.55, p < 0.001$), further confirming its superior diagnostic performance in predicting the necessity of revascularization.

3.6. Correlation Between cHSR, TAWSS, and OSI

The relationship between cHSR and the shear stress parameters TAWSS and OSI was evaluated. No significant correlation was found between cHSR and TAWSS ($r = -0.05, p = 0.61$), nor between cHSR and OSI ($r = 0.14, p = 0.16$) (Figure 6).

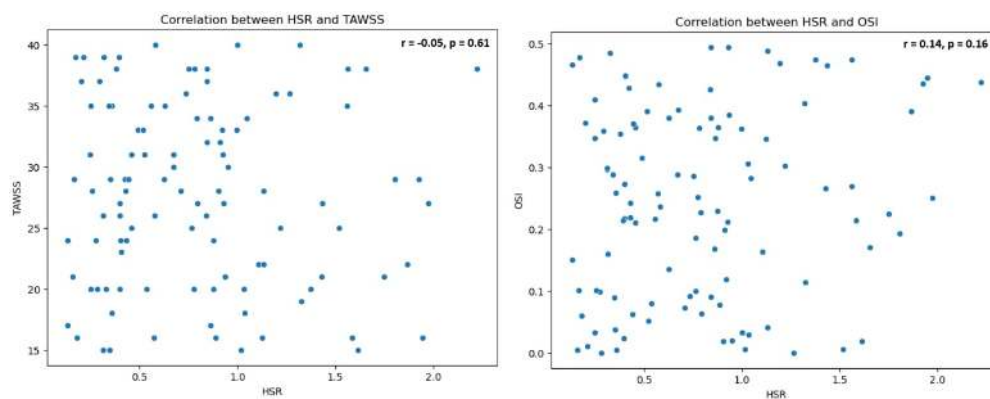


Figure 6. Scatter plots for HSR vs. TAWSS and HSR vs. OSI: HSR: Hyperemic Stenosis Resistance; TAWSS: Time-Averaged Wall Shear Stress; OSI: Oscillatory Shear Index.

4. Discussion

4.1. Study Population

Our study included a cohort with suspected chronic coronary syndrome (CCS), encompassing a broad range of pretest probabilities and coronary lesion complexities. This diversity underscores the applicability of coronary computed tomography angiography (CCTA) as an initial diagnostic tool across varied patient profiles, aligning with recent guidelines and recommendations [2]. In a metaanalysis conducted by Foy and al. [23], although CCTA did not significantly reduce mortality or cardiac hospitalizations, it was associated with a lower incidence of myocardial infarction, alongside an increase in invasive angiography, revascularizations, coronary artery disease diagnoses, and new prescriptions for aspirin and statins, compared to functional stress testing. This finding suggests CCTA's role in identifying a higher burden of clinically significant CAD. Thus, traditional functional imaging—while valuable for ischemia detection—may have limited diagnostic performance in specific populations, such as those with three vessel disease and left main lesions, where anatomical assessment may offer superior guidance [12].

4.2. FFR_{CFD} and FFR Correlation and Lumped Model Validation

In our study, FFR_{CFD} demonstrated excellent agreement with invasive FFR, with a Pearson correlation coefficient of 0.87, which is comparable to the results by Nørgaard et al. [24], who reported a correlation coefficient of 0.81, and Faulder et al. [25], who noted values between 0.82 and 0.85 in large, multicentric cohorts. Bland-Altman analysis in our study showed minimal bias, with most differences falling within ± 0.10 , affirming our model's accuracy (Figure 7).

However, discrepancies between FFR_{CFD} and invasive FFR were observed in four cases, highlighting areas where further refinement of computational models may be beneficial. Two cases involved patients with a silent myocardial infarction (MI) in the territory of the coronary lesion. Prior research has shown that FFR values in post-MI patients can vary depending on the extent of myocardial viability. De Bruyne et al. demonstrated that viable myocardial tissue can maintain vasoreactivity even in infarcted regions, leading to variability in FFR measurements due to altered blood flow dynamics [26]. This suggests that in cases of silent MI, adjustments to computational models may be needed to accurately account for myocardial viability and regional perfusion changes.

Two other cases involved patients with a high coronary calcium burden, where segmentation challenges affected the precision of CCTA-derived models. High calcium scores can impair the accuracy of CCTA, impacting the segmentation process and introducing potential errors in model creation. Quality CCTA acquisition and advanced

segmentation algorithms are critical in such scenarios, while the use of ultrahigh-resolution photon-counting CT may resolve this problem, with resolution levels approaching those of invasive coronary angiography [27].

The model's high correlation likely benefits from the rigorous tuning of boundary conditions and patient-specific inputs such as flow adjusted to myocardial mass, which further improved alignment with invasive measures in respect to methods described previously [28,29].

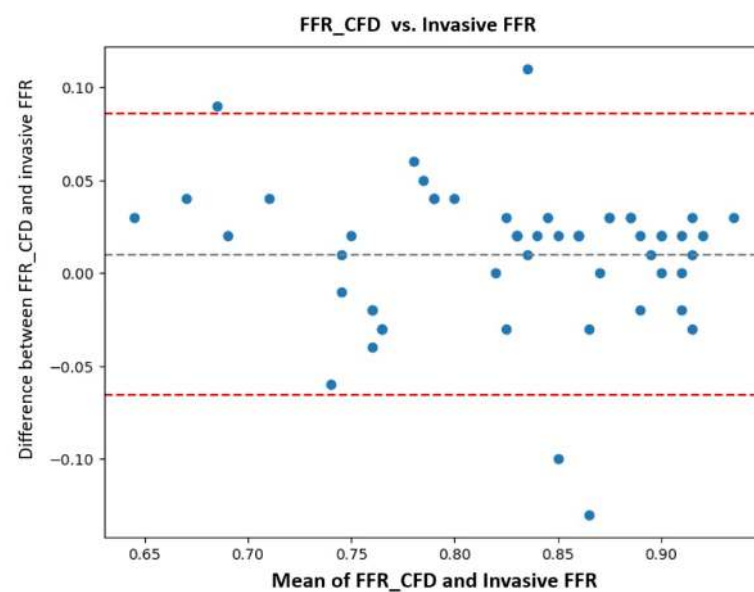


Figure 7. Bland-Altman analysis FFR_{CFD} vs. invasive FFR: computationally derived fractional flow reserve (FFR_{CFD}); invasive fractional flow reserve (FFR).

4.3. *cHSR Cutoff*

Our analysis identified an optimal *cHSR* cutoff of $0.75 \text{ mmHg/cm}\cdot\text{s}^{-1}$ for predicting the necessity of revascularization, aligning closely with the findings of Meuwissen et al. [9], who proposed a hyperemic stenosis resistance (HSR) threshold of $0.8 \text{ mmHg/cm}\cdot\text{s}^{-1}$ in invasive measurements. However, this slight adjustment in cutoff in our study may result from the fidelity of our simulation methods. Despite promising preliminary evidence, larger studies are needed to substantiate this finding, especially to achieve the statistical power required to define an optimal *cHSR* threshold across diverse populations. Thus, prospective validations are crucial to reinforce the clinical applicability of *cHSR* as a complementary measure to FFR_{CFD} .

4.4. Diagnostic Performances of HSR, QFR, and FFR_{CFD}

Both FFR_{CFD} and QFR displayed high diagnostic performance, consistent with prior studies (Table 4) such as Xu et al. [14]. and Agatssthi et al. [30], who reported sensitivities and specificities exceeding 80%. In our study, FFR_{CFD} and QFR achieved accuracies of 83% and 81%, respectively. However, cHSR demonstrated superior performance, with an accuracy of 96%, along with higher sensitivity and specificity than both FFR_{CFD} and QFR. This enhanced diagnostic capability of cHSR aligns with the conclusions of Van de Hoef et al. [31], who noted that combined pressure and flow measurements can provide a more comprehensive understanding of coronary physiology than pressure indices alone. This combined approach for HSR appears to capture the hemodynamic impact of coronary stenosis more effectively, potentially refining patient selection for revascularization and addressing limitations inherent to pressure-only indices.

Table 4. Summary of key studies evaluating the diagnostic performance and clinical benefit of computational fluid dynamics (CFD) and imaging-based coronary physiology indices.

Study	Index	Sample Size	Modality	Key Findings
Nørgaard et al., 2014 [24]	FFR_CT	254	CCTA	Demonstrated high diagnostic accuracy of FFR_CT with invasive FFR as reference, sensitivity and specificity >80%.
Westra et al., 2018 [34]	QFR	352	Angiography	QFR was highly accurate in predicting ischemia with FFR reference; achieved >85% accuracy.
Xu et al., 2021 [14] (FAVOR III China)	QFR	3825	Angiography	Showed significant reduction in MACE over 1 year for QFR-guided PCI vs. standard angiography-guided PCI.
Song et al., 2022 [35] (2-Year FAVOR III)	QFR	3825	Angiography	Demonstrated durability of QFR-guided PCI benefits with reduced MACE over 2 years compared to angiography guidance.
Driessen et al., 2019 [36]	FFR_CT	208	CCTA	FFRCT showed higher diagnostic performance than standard coronary CTA, SPECT, and PET for vessel-specific ischemia.
Koo et al., 2011 [37]	FFR_CT	103	CCTA	Accuracy, sensitivity, specificity, positive predictive value, and negative predictive value were 84.3%, 87.9%, 82.2%, 73.9%, 92.2%, respectively, for FFR_CT.

FFRCT: Fractional Flow Reserve derived from Coronary Computed Tomography Angiography; QFR: Quantitative Flow Ratio; CCTA: Coronary Computed Tomography Angiography; PCI: Percutaneous Coronary Intervention; MACE: Major Adverse Cardiac Events; SPECT: Single Photon Emission Computed Tomography; PET: Positron Emission Tomography.

4.5. HSR and Shear Stress

Our investigation revealed no significant correlation between cHSR and shear stress parameters such as TAWSS and OSI, suggesting rather an intricate relationship between cHSR and shear stress. The literature on this topic is limited, particularly for computational indices, but some studies on invasive measures provide context. For instance, Puri et al. reported that pathological shear stress correlated with plaque vulnerability [32], while Wong et al. found moderate relationships between invasive indices and WSS parameters in specific lesion types [33]. The absence of a significant relationship in our study may indicate that while cHSR captures functional stenosis effects, it may not reflect local shear stress changes that influence plaque morphology and stability. This finding highlights the complexity of hemodynamic interactions, and the insufficient character of the current models to fully understand them.

5. Limitations

This study has several limitations. First, the retrospective design introduces potential selection biases, and the relatively small sample size may limit the generalizability of findings to larger populations. Additionally, the computational power and time required for 3D flow simulations pose challenges, as these processes are resource consuming and not yet feasible for routine clinical application. The study relied on retrospective acquisition protocols for CCTA, which increased cumulative radiation exposure for patients. Furthermore, the simulation models assume rigid arterial walls and omit arterial motion, which may not fully represent the dynamic nature of coronary arteries *in vivo*. Future work with larger sample sizes and advanced computational models may help overcome these limitations and increase the applicability of computational hyperemic stenosis resistance (cHSR) as a reliable diagnostic tool.

6. Conclusions

Our study (Figure 8: Central illustration) demonstrates the promising diagnostic potential of a new computational index (cHSR), showing superior performance compared to traditional imaging-based indices. The computational model used in this study produced values of FFR_{CFD} with strong agreement with invasive FFR. These findings support the integration of computational methods into non-invasive imaging to improve decision-making in CCS management. Future advancements in automation and machine learning could enhance its clinical viability, potentially transforming how CCS is assessed and treated.

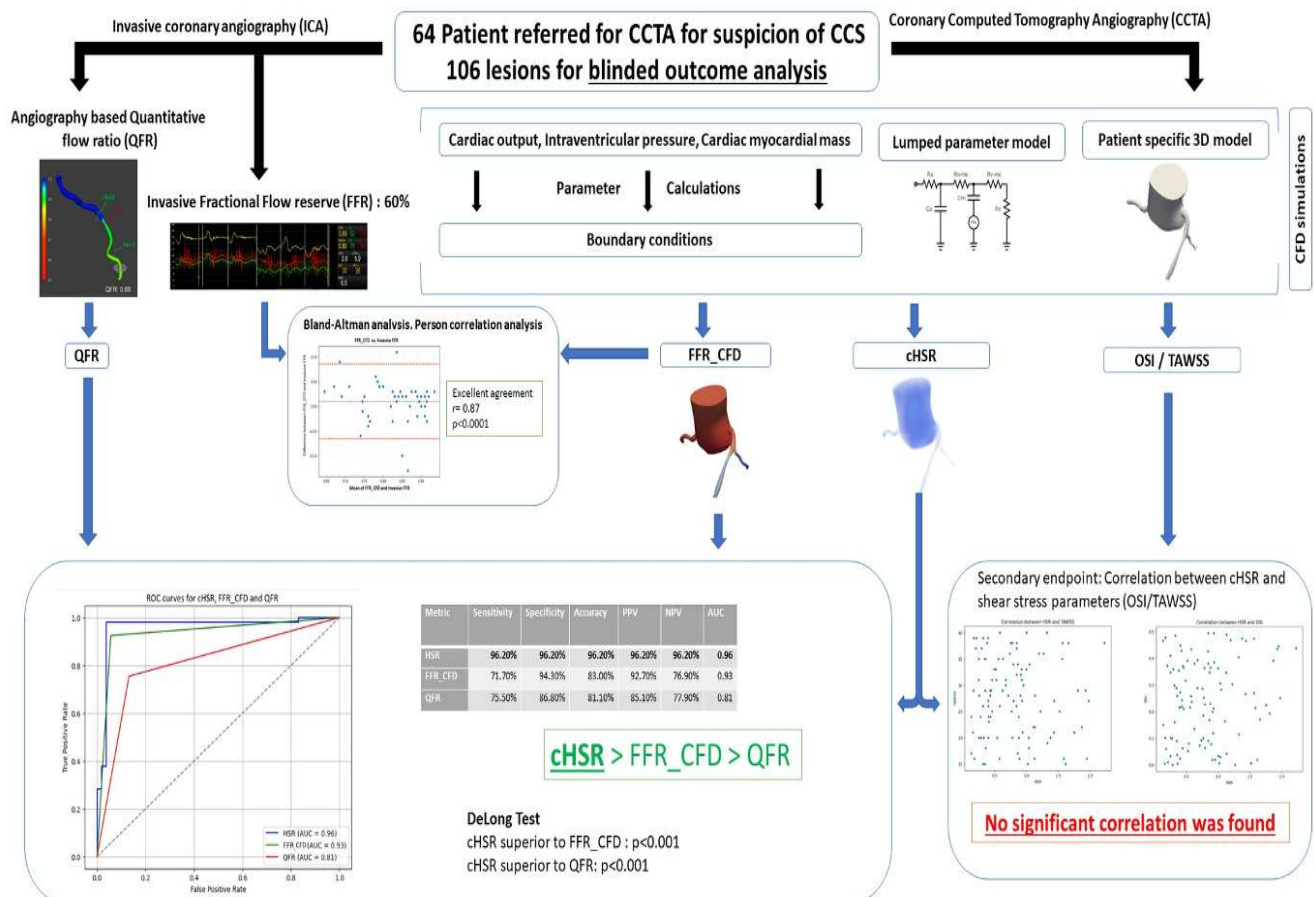


Figure 8. Central illustration: CCS: Chronic Coronary Syndrome; CCTA: Coronary Computed Tomography Angiography; ICA: Invasive Coronary Angiography; QFR: Quantitative Flow Ratio; FFR: Fractional Flow Reserve; FFR_{CFD}: Fractional Flow Reserve derived from Computational Fluid Dynamics; cHSR: Computational Hyperemic Stenosis Resistance; OSI: Oscillatory Shear Index; TAWSS: Time-Averaged Wall Shear Stress; CFD: Computational Fluid Dynamics; AUC: Area Under the Curve; PPV: Positive Predictive Value; NPV: Negative Predictive Value.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/doi/s1>.

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Institutional Review Board Statement: The Brest university hospital ethics board approved the study protocol which was based on Reference Methodology (MR004), as defined by the French Data Protection Authority (CNIL), and patient data were anonymized in compliance with institutional regulations (Approval date: 24 September 2024).

Informed Consent Statement: Informed consent was waived for this retrospective study in accordance with Reference Methodology MR004 and approval by the Brest University Hospital ethics board.

Data Availability Statement: The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request. Due to patient confidentiality and ethical restrictions imposed by the Institutional Review Board, the raw data cannot be publicly shared. However, de-identified data relevant to the findings of this study may be provided upon request, subject to institutional approval and compliance with data protection regulations.

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Conflicts of Interest: The authors declare no conflict of interest.

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Supplementary Methods

Patient-Specific 3D Model Elaboration

The 3D model construction for this study focused on representing the ascending aorta and the main branches of the coronary tree. Each coronary artery was modelled up to its first major bifurcation (except for the left main bifurcation giving anterior descending and left circumflex). All arteries extended for additional 15 mm beyond a lesion if possible.

Segmentation Process

Segmentation was performed using an automatic machine learning (ML) algorithm integrated with SimVascular¹. This segmentation approach allowed accurate identification of the coronary lumen, even in cases with high calcification. The segmentation process began with the selection of paths along each vessel's centerline. Paths were created using the “Control Points” feature in SimVascular, enabling alignment with the center of the coronary lumen. Paths were smoothed using Fourier smoothing to ensure a straight, consistent course through each centerline. Following path planning, 2D segmentations were generated along each path and then lofted to create the vessel geometry.

Meshing and Element Sizing

A radius-based meshing approach was applied to each vessel, with element sizes set at 0.03 mm to allow for a finer mesh along smaller vessels and complex curved segments. This method ensured sufficient detailing, particularly near regions of bifurcation and stenosis. By adhering to these meshing protocols, the final 3D model achieved the necessary balance between computational efficiency and accuracy in simulating coronary blood flow. A meshing independence test was realized to ensure less than 1.2% of the relative pressure fields cyclic variation between cycles at the inlet and outlets.

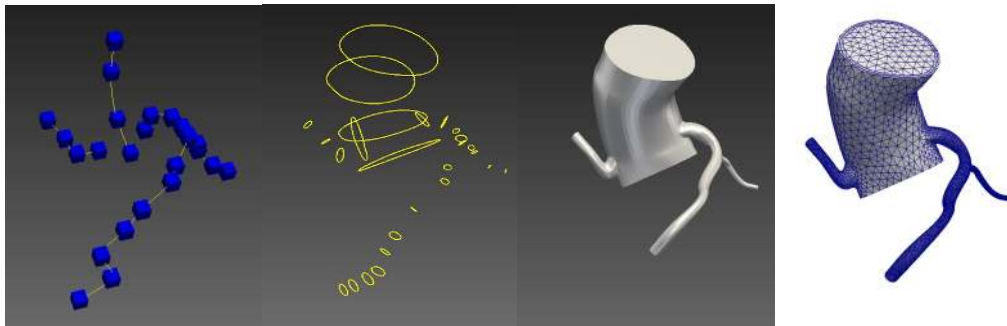


Figure S1. Steps to create a patient-specific 3D model of the vascular system. From left to right: (1) Extraction of vessel centerlines from imaging data, providing a simplified representation of vascular geometry. (2) Segmentation using machine learning algorithms to identify and delineate vessel boundaries. (3) Lofting process to construct a smooth 3D model of the vessels based on the segmented cross-sections. (4) Meshing of the 3D model for computational simulations, ensuring proper resolution and accuracy for analysis.

Simulation model parameters

Input parameters sources

To achieve accurate and patient-specific simulations, we gathered essential input data from multiple sources. Each parameter was tailored to match the individual patient's physiological characteristics using both non-invasive and invasive measurements.

- **Aortic inflow rate** was derived from MRI phase contrast data of a reference subject with similar Body Surface Area (BSA) to the patient. This reference flow profile was scaled to fit each patient's stroke volume and systolic ejection time, which we extracted from Cardiac CT volumetry measures. Myocardial mass was also calculated accordingly, and a perfusion percentage was allocated to each artery based on visual inspection and coronary dominance.
- **Patient-specific systolic and diastolic pressures** were obtained from invasive blood pressure measurements. These values provide the necessary inputs for modeling arterial pressure at various stages of the cardiac cycle. Noninvasive measurements can be a viable alternative in most of cases.
- **Intramyocardial pressure (P_{im})** was modeled using an invasive left ventricular pressure curve obtained from healthy subjects. To adapt this reference pressure curve to each patient, we scaled it according to the patient's specific aortic pressure measurements. This scaling ensures that P_{im} reflects the patient's myocardial loading conditions while preserving the temporal profile of left ventricular pressure changes. For modelling the right coronary arteries P_{im} , a scaling of 0.33-0.4 was applied to simulate the lower pressure levels in the right ventricle.

Aortic Outflow Model

For the aortic outflow, a classic RCR Windkessel model was considered, which incorporates three elements: proximal resistance (R_p), compliance (C), and distal resistance (R_d). These components collectively represent the systemic arterial system's resistance and elasticity. At baseline, we calculated the total systemic resistance from the patient's mean arterial pressure and cardiac output. The total resistance was distributed into R_p and R_d in a ratio of 1:9, emphasizing the higher resistance contribution of the distal microcirculatory beds. Compliance was initially set to 0.001 cm⁵/dyne, ensuring a physiologically realistic systolic-diastolic pressure waveform with an amplitude in a reasonable range (40-60 mmHg).

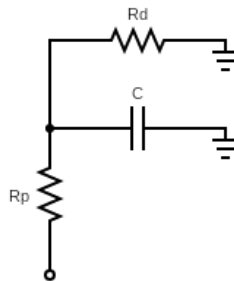


Figure S2. Schematic representation of the RCR (Resistance-Capacitance-Resistance) Windkessel model, showing proximal resistance (R_p), distal resistance (R_d), and compliance (C) used to simulate vascular impedance and flow dynamics.

Under stress conditions, cardiac output was increased by 300-350%, achieved through a combination of a 200-250% rise in stroke volume and a 120-160% increase in heart rate. The raised cardiac output necessitated recalibration of systemic resistance, targeting an increase by 140-150% of the mean aortic pressure. To maintain realistic pulsatile pressure waveforms, aortic compliance was reduced by 60% to reflect the stiffer arterial conditions seen in stress. A second process of tuning in 5% steps was performed to ensure a smooth pressure decay and reasonable systolic-diastolic pressure differences. These adaptations allowed the aortic outflow model to accurately respect the hemodynamic shifts associated with increased workload.

Equation S1

$$R_{tot} = \frac{\text{Mean pressure}}{\text{Cardiac output}}$$

Lumped Parameter Model for Coronary Outflows

The coronary circulation was represented using a lumped parameter network ² comprising resistances (R_a , $R_{a\text{-micro}}$, R_v), capacitances (C_a , C_{im}), and an intramyocardial pressure (P_{im}):

- **Resistances (R_a , $R_{a\text{-micro}}$, R_v):** These represent the flow impedances in different regions of the coronary vasculature. R_a models proximal resistance in larger arteries, $R_{a\text{-micro}}$ corresponds to resistance within the microcirculation, and R_v accounts for resistance in the venous outflow. We considered a zero value of $R_{v\text{-micro}}$, which had little effect on the output as reported previously ³.
- **Capacitances (C_a , C_{im}):** These simulate the ability of vessels to store blood. C_a models arterial compliance, while C_{im} reflects the compliance within the myocardium, influenced by myocardial contraction. A ratio of 0.11:0.89 was constantly used in our simulations, as previously reported ³.
- **Intramyocardial Pressure (P_{im}):** This pressure, derived from left ventricular pressure curves, captures the compressive forces exerted on coronary vessels during systole and diastole. It introduces a key feature of coronary physiology: the out-of-phase relationship where flow is predominantly diastolic.

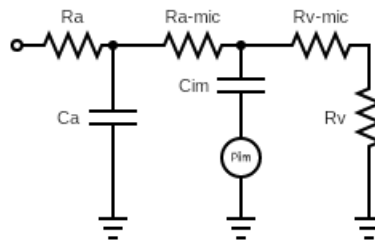


Figure S3 Lumped parameter model of coronary circulation, depicting proximal arterial resistance (R_a), microvascular arterial resistance ($R_{a\text{-mic}}$), microvascular venous resistance ($R_{v\text{-mic}}$), venous resistance (R_v), arterial compliance (C_a), microvascular compliance

This model was able to reproduce the unique out-of-phase flow characteristics of coronary arteries, where flow is predominantly diastolic. The total coronary flow was set at 4-5% of cardiac output, consistent with physiological coronary perfusion. The flow distribution to each artery was adjusted to the patient's myocardial mass based on previous studies ⁴⁻⁸. Total coronary resistance (using the same equation as

equation 1 for coronary circulation) was found as 24 times the systemic resistance to account for the higher resistance characteristic of coronary microcirculation. At rest, this resistance was distributed among the components as follows: R_a (32%), $R_{a\text{-micro}}$ (52%), and R_v (16%) as reported in previous studies^{3,9}. Intramyocardial pressure (P_{im}), derived from left ventricular pressure curves, was scaled to match patient-specific pressures. A scaling of 0.33 of P_{im} for the right coronary arteries (RCA) was respected, ensuring accurate external compression by the right ventricle.

The distribution of coronary resistance and compliance among outlets was guided by physiological principles and geometric considerations. Resistance was allocated based on the outlet's distal cross-sectional area following a generalization of Murray's Law, which posits that resistance is inversely proportional to the radius raised to the power of 2.6. Similarly, compliance distribution was proportionate to the outlet area.

Equation S2

$$R_i = R_{\text{total}} \cdot \frac{r_i^{-2.6}}{\sum_j r_j^{-2.6}}$$

Where:

- R_i is the resistance assigned to the outlet i .
- R_{total} is the total coronary resistance.
- r_i is the radius of the outlet.
- The denominator $\sum_j r_j^{-2.6}$ is the summation of all outlet radii raised to the power of -2.6

Equation S3

$$C_i = C_{\text{total}} \cdot \frac{A_i}{\sum_j A_j}$$

Where:

- C_i is the compliance assigned to the outlet i .
- C_{total} is the total coronary compliance.
- A_i is the cross-sectional area of outlet i .

Under stress, the coronary model underwent several adjustments to accommodate increased cardiac workload. While the total coronary flow maintained its proportion of 4-5% of the elevated cardiac output, the total coronary resistance was recalculated to align with the higher cardiac output and redistributed among the resistance components (R_a , $R_{a\text{-micro}}$, R_v) using a different ratio (45%, 33%, 22%). This distribution strategy preserved coronary flow in regions with non-significant (<30%) or no stenosis while recreating the physiological pressure drop observed in the presence of a more significant one. Coronary compliance was increased by 100% to accommodate the higher coronary flow demands, and tuned in 10% steps to achieve smooth pressure curves. Additionally, P_{im} was scaled to higher levels to match increase in systolic aortic

pressure. Distribution of resistance and compliance to each coronary outlet respected the same reasoning as at rest.

Table S1. Initial values and tuning process

PARAMETER	INITIAL VALUE OR CALCULATION METHOD	STRESS ADJUSTMENTS	TUNING PROCESS/TARGET
AORTIC RESISTANCE (R)	Calculated using mean aortic pressure and cardiac output, split as Rp: Rd in 1:9 ratio.	Recalculated to reflect 140-150% increase in mean aortic pressure.	Automatically scaled with cardiac output to match increased systemic pressures.
AORTIC COMPLIANCE (C)	Set to 0.001 cm ⁵ /dyne to match physiological systolic-diastolic pressure waveforms.	Reduced by 60% to account for arterial stiffening during stress.	Adjusted iteratively in 5% steps to maintain reasonable pressure decay and systolic-diastolic amplitude.
TOTAL CORONARY FLOW	Fixed at 4-5% of cardiac output, normalized to myocardial mass.	Increased proportionally with cardiac output to maintain 4-5% coronary flow distribution under stress.	No additional tuning; scaled directly with cardiac output.
TOTAL CORONARY RESISTANCE (R _{COR})	Calculated as 24x systemic resistance, distributed among Ra (32%), Ra-micro (52%), and Rv (16%).	Recalculated based on increased cardiac output, distributed among Ra (45%), Ra-micro (33%), and Rv (22%).	Adjusted to achieve accurate pressure-flow curves.
R _A (PROXIMAL RESISTANCE)	32% of total coronary resistance, calculated based on outlet diameters using Murray's Law ($r^{2.6}$).	45% of total coronary resistance; adjusted to ensure physiological stress flow drop in significant lesions	Adjusted to achieve accurate pressure-flow curves.
R _{A-MICRO} (MICROCIRCULATORY RESISTANCE)	52% of total coronary resistance, reflecting capillary network characteristics.	33% of total coronary resistance	Adjusted to achieve accurate pressure-flow curves.
R _V (VENOUS RESISTANCE)	16% of total coronary resistance, representing venous return properties.	22% of total coronary resistance	Adjusted to achieve accurate pressure-flow curves.
CORONARY COMPLIANCE (C)	Distributed as C _a (11%) and C _{im} (89%) based on outlet area.	Increased by 100% to accommodate higher coronary flows under stress conditions.	Adjusted iteratively in 10% steps to match physiological pressure curves at each coronary outlet.
INTRAMYOCARDIAL PRESSURE (PIM)	Derived from left ventricular pressure curve, scaled 0.33 for RCA.	Scaled to meet aortic systolic pressure increase	No further tuning

Reduced order models and tuning process

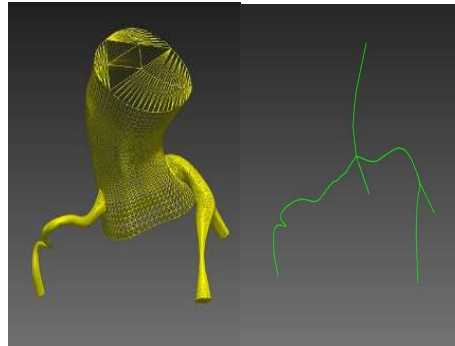


Figure S4. From 3D model to 1D centreline for reduced order simulations

The 1D model is generated by extracting vascular centerlines from a 3D model, discretized into segments for finite element analysis of flow and pressure. Lumped parameter models represent the aortic and coronary trees using resistances, compliances, and intramyocardial pressure. Iterative simulations are conducted to tune these parameters, including resistance distributions and compliance values, through trial-and-error adjustments, ensuring alignment with defined boundary conditions and flow dynamics (Table S1. Initial values and tuning process).

Solver parameters description

In simulations, wall properties assume rigid vessel walls with no-slip and no-penetration boundary conditions applied to blood flow. While this approach simplifies the model, more advanced simulations can incorporate wall deformation through fluid-structure interaction (FSI) for greater realism, this approach in the other hand, increase computational resources necessary to perform simulations. Solver parameters were configured to ensure computational stability. The simulations were set to run for 4000-9000 timesteps with a timestep size of 0.001 seconds, adjusted to the patient heart rate of beats per minute (cardiac cycle duration in seconds). Results were saved every 20 timesteps, with linear solver settings (Step Construction = 5 and Maximum Iterations = 10) optimized for coronary simulations. Blood was modeled with a viscosity of 0.04 poise and a density of 1.06 g/cm^3 , which are standard physiological values. These settings ensure the fidelity of the hemodynamic simulation while maintaining computational efficiency.

Supplementary appendix references

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